

Md. Jahidul Hoque

+33 751044629 — mdjahidul.hoque@agroparistech.fr — mdjahidulhoque — mdjahidulhoque.github.io — Paris, France

Education

AgroParisTech - Université Paris-Saclay, France

Master of Science (M2) in Climate, Land-Use and Ecosystem Services

Sep 2024 - Oct 2025
Mention: Good - Ranked 3rd out of 13

Bangladesh Agricultural University (BAU), Bangladesh

Master of Science (MSc) in Environmental Science

Oct 2022 - Mar 2024

CGPA: 3.92/4.00

Noakhali Science and Technology University (NSTU), Bangladesh

Bachelor of Science (BSc) in Environmental Science

Jan 2017 - Jul 2022

CGPA: 3.64/4.00

Professional Experience

Research Intern

LSCE (Laboratoire des Sciences du Climat et de l'Environnement; CEA-CNRS), MOSAIC Team

May 2025 - Oct 2025

Gif-sur-Yvette, France

- Contributed to the implementation and evaluation of the Johnson & Berry (2021) mechanistic photosynthesis model within the ORCHIDEE land-surface model.
- Participated in early-stage integration of mechanistic photosynthesis formulations in a terrestrial biosphere model (TBM), supporting fluorescence-constrained analyses.
- Performed literature synthesis and dataset compilation on Pulse-Amplitude Modulation (PAM) chlorophyll fluorescence.
- Conducted quantitative meta-analysis of photosynthesis and fluorescence metrics to identify variability and data gaps across ecosystems.

Research Assistant

Dr. Murad Ahmed Farukh's Lab, Bangladesh Agricultural University

Nov 2022 - Apr 2024

Mymensingh, Bangladesh

- Worked on an FAO-funded project on agricultural land transformation in floodplain systems of Bangladesh.
- Analysed land-use/land-cover dynamics using Google Earth Engine and satellite imagery within a GIS framework.
- Conducted field surveys and integrated ground observations with remote-sensing products for validation.

Research Assistant

Department of Environmental Science and Disaster Management, NSTU

Jan 2020 - Jul 2022

Noakhali, Bangladesh

- Supported research on environmental monitoring, soil and water quality, and ecological assessment using field, laboratory, and GIS-based approaches.
- Conducted remote sensing and geospatial analyses to support environmental studies.
- Trained undergraduate students in programming and statistical analysis for environmental data interpretation.

Research Outputs

Peer-Reviewed Publications

- Hoque, M. J., Hossain, M. S., Sejuti, S. I., & Farukh, M. A. (2026). Transforming the Floodplain: Agricultural Encroachment and Landscape Change in the Northwestern Ganges-Padma Basin. *Geo: Geography and Environment*, 13:e70067. DOI: 10.1002/geo2.70067. (Derived from Master's thesis, Bangladesh Agricultural University, 2024).
- Farukh, M. A., Brahma, P. P., Hossain, M. S., Hoque, M. J., Sejuti, S. I., Shammy, U. S., & Arefin, K. S. (2026). Climatological assessment of pre-monsoon heatwave days in Bangladesh and their relationship to Indo-Pacific circulation anomalies. *Natural Hazards*, 122:32. DOI: 10.1007/s11069-025-07746-7.

Manuscripts Under Review

- Hossain, M. S., Farukh, M. A., Hoque, M. J., Brahma, P. P., & Arefin, K. S. (2026). Coupling Land Use Change, Surface Warming, and Methane Emissions in Bangladesh's Haor Wetlands: A 33-Year Remote Sensing Assessment (1990–2023).

Master's Thesis (M2)

- Enhancing Global Photosynthesis Estimation through the Farquhar Model Recharged: Mechanistic Light-Dependent Reactions of Photosynthesis. *AgroParisTech - Université Paris-Saclay, conducted at LSCE (CEA-CNRS), 2025.*

Conference Contributions

- Martini, D., Hoque, M. J., Johnson, J. E., Wohlfahrt, G., & Maignan, F. (2025). Temperature Response of the Johnson & Berry Model: Steps toward SIF-Constrained GPP Estimation in ORCHIDEE. *AGU 2025 (Poster)*.
- Hoque, M. J., Hossain, M. S., & Farukh, M. A. (2024). Assessment of the Transformation of Agricultural Land Based on Land Use and Land Cover (LULC) Dynamics in the Ganges-Padma Floodplain. *International Conference on Advanced Agricultural Research, Sylhet Agricultural University, Bangladesh.*
- Hossain, M. S., Hoque, M. J., & Farukh, M. A. (2024). Assessment of Land Use Land Cover (LULC) and Land Surface Temperature (LST) on Methane (CH₄) Concentration Emitted from Agricultural Fields in the North-Eastern Haor Areas of Bangladesh. *International Conference on Advanced Agricultural Research, Sylhet Agricultural University, Bangladesh.*

Skills

- **Programming and Data Analysis:** Python, R, Fortran, SPSS, Minitab
- **GIS and Remote Sensing:** Google Earth Engine, ArcGIS, ArcGIS Pro, QGIS, ILWIS
- **Environmental Modelling and Scientific Tools:** ORCHIDEE, SimaPro, OriginPro, GraphPad Prism
- **Languages:** Bengali (Native), English (C1), French (basic; currently improving)

Conferences

- ICOS Science Conference 2024, Versailles Palais des Congrès, France (Attended)
- International Conference on Advanced Agricultural Research, Sylhet Agricultural University, Bangladesh (Presented findings), May 2024
- 8th Annual Conference on Social Research in Bangladesh, Bangladesh Institute of Social Research Trust (Presented findings), November 2023
- National Conference on Resource Management Sustainability and Climate Change, Department of Geography and Environment, Dhaka University, Bangladesh (Attended), March 2019
- 7th International Conference on Water and Flood Management (ICWFM), Focusing on Water Security Under a Changing Climate (Attended), March 2019

Training and Courses

- Training on Statistical Packages (SPSS, R, Minitab), Department of Agrometeorology, Bangladesh Agricultural University, October 2023
- Training on urban tree inventory using i-Tree, United States Forest Service, August 2023
- Training on Application of Google Earth Engine (GEE) in Forestry, US Forest Service and USAID at Khulna University, May 2023
- Training on R Programming for Research and Data Analysis, Research Learning Counter, Bangladesh, February-March 2022
- Training on Environmental Science and Engineering and Sustainability using Life Cycle Assessment, Dr. Nawshad Haque, Senior Scientist (CSIRO Australia), September-December 2020
- MOOC (Massive Open Online Course) - Cartography facilitated by ESRI, April 2023
- MOOC - Going Places with Spatial Analysis facilitated by ESRI, March 2023

Honours and Awards

- French Government Scholarship (Boursier du Gouvernement Français - BGF) for M2 studies at AgroParisTech - Université Paris-Saclay
- Eiffel Excellence Scholarship (Master's level) - Nominated by Université Paris-Saclay
- National Science and Technology (NST) Fellowship, Bangladesh
- Education Board Scholarships during Junior School Certificate (JSC), Secondary School Certificate (SSC), and Higher Secondary Certificate (HSC)

Extra-Curricular Activities

NSTU Photography Club <i>President</i>	Jan 2021 - Aug 2022
NSTU Adventure Club <i>Joint Secretary</i>	Jan 2021 - Dec 2021
Dhrupod Musical Club - NSTU <i>Joint Documentation Secretary</i>	Apr 2020 - Mar 2022
Coastal Environmental Network (CoEN) <i>Organizing Secretary</i>	Feb 2018 - Feb 2019